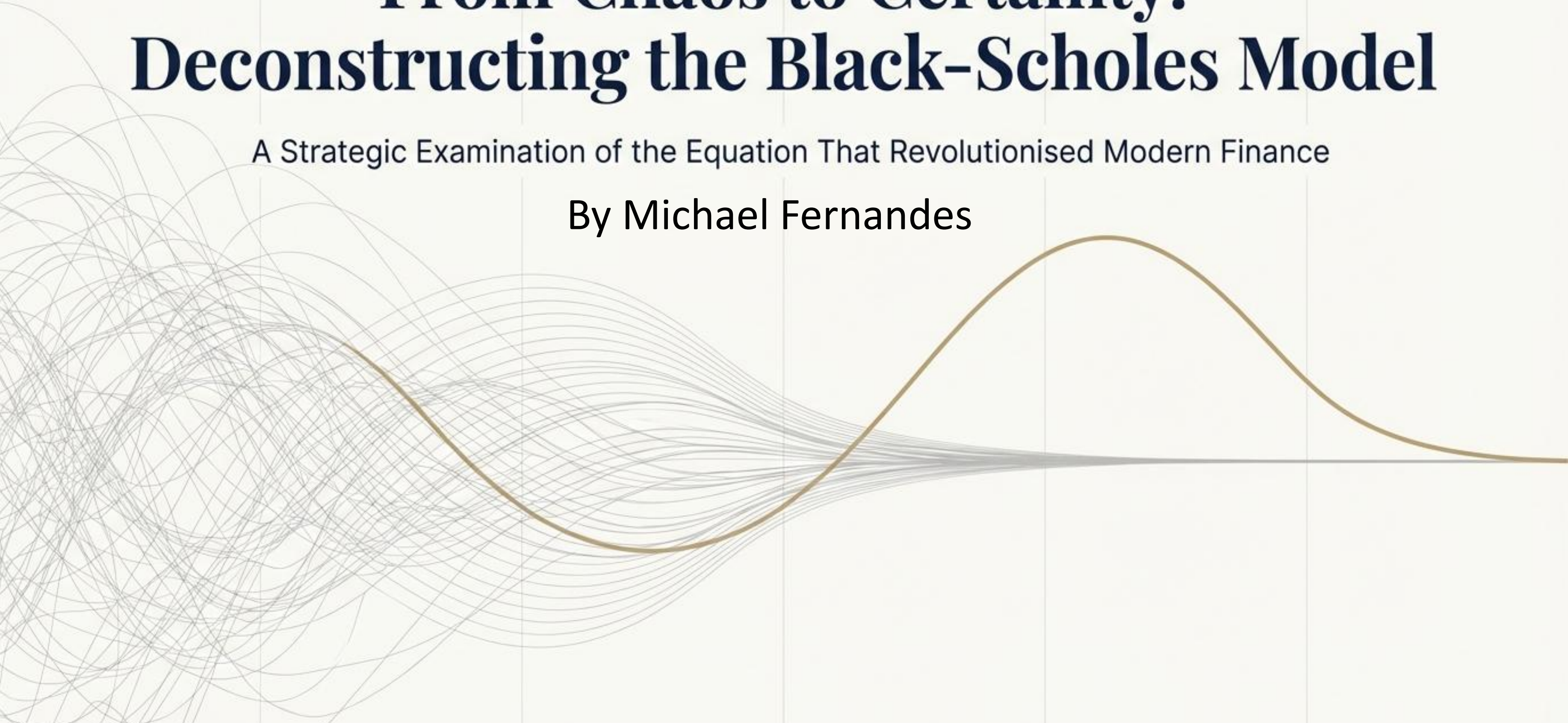


From Chaos to Certainty: Deconstructing the Black-Scholes Model

A Strategic Examination of the Equation That Revolutionised Modern Finance

By Michael Fernandes



The Black-Scholes Model at a Glance

Core Concept: How it Works



The No-Arbitrage Principle

The model finds an option's single correct price by creating a perfectly hedged portfolio.



The 5 Key Inputs



Stock Price (S)
The current market price of the underlying asset.



Strike Price (X)
The price at which the option can be exercised.



Time to Expiration (T)
The remaining lifespan of the option, in years.



Risk-Free Rate (r)
The interest rate of a risk-free investment, in years.



Volatility (σ)
The standard deviation of the stock's returns.

Practical Use: Managing Risk with "The Greeks"

Measuring Risk Sensitivities

The "Greeks" are partial derivatives of the model that tell traders how an option's price will change.

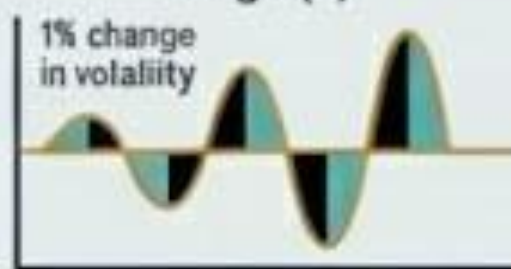
Delta (Δ)



Measures Sensitivity To...
Stock Price

What it Tells You:
How much the option's price moves for a £1 change in the stock price.

Vega (v)



Measures Sensitivity To...
Volatility

What it Tells You:
How much the option's price moves for a 1% change in volatility.

Theta (θ)



Measures Sensitivity To...
Time Decay

What it Tells You:
How much the option's price moves each day as it nears expiration.

The Fundamental Insight: A Perfectly Hedged World

Under the model's assumptions, an option can be perfectly and continuously hedged by buying and selling the underlying stock. This process eliminates all market risk from the position.

The Consequence

- “No Randomness”: The portfolio's value is no longer subject to random market shocks.
- “No Subjectivity”: Its return is independent of investors' views on the stock's future performance.

Because the position is risk-free, it must earn the risk-free interest rate. This principle of no-arbitrage dictates that there is only one correct price for the option.

Underlying
Stock (S)



+

Option (V)



=

Hedged
Portfolio (Π)



The Destination: The Black-Scholes Equation

The diagram shows the Black-Scholes equation with callouts for its components:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Callouts:

- Theta (Θ) - Time Decay (points to $\frac{\partial V}{\partial t}$)
- Delta (Δ) - Price Sensitivity (points to $\frac{\partial V}{\partial S}$)
- Gamma (Γ) - Delta's Sensitivity (points to $\frac{\partial^2 V}{\partial S^2}$)

V: The price of the derivative (our goal).

S: The price of the underlying stock.

t: Time.

r: The risk-free interest rate.

σ : The volatility of the stock's returns.

The Axioms: Establishing a Frictionless Universe

The model's logic holds true within a specific set of idealised conditions.



Stock Price Behaviour

Follows a Geometric Brownian Motion with constant volatility (σ) and drift (μ). Returns are log-normally distributed.



Trading Environment

Trading is continuous and frictionless (no transaction costs or taxes). Short selling is permitted.



Market Conditions

The risk-free rate (r) is constant. There are no arbitrage opportunities.



Option Characteristics

The option is European (exercisable only at expiration) and the underlying stock pays no dividends.

Derivation Step 1: Modelling the Random Walk

We begin by assuming the stock price follows a process known as Geometric Brownian Motion.

$$dS = \mu S dt + \sigma S dW$$



$\mu S dt$ (The Drift):

A deterministic term representing the expected return or trend of the stock over time. μ is the expected rate of return.

$\sigma S dW$ (The Diffusion):

A stochastic term representing the random, unpredictable volatility. dW is the Wiener process increment—the source of randomness.

Our entire goal is to build a portfolio where the random dW term is completely eliminated.

Derivation Step 2 & 3: The Moment of Genius - The Riskless Portfolio

Itô's Lemma

First, we need to know how the option price, $V(S,t)$, changes. Itô's Lemma gives us the answer:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \left(\frac{\partial^2 V}{\partial S^2} \right) \right) dt + \underbrace{\sigma S \frac{\partial V}{\partial S} dW}$$

The Hedge

We construct a specific portfolio (Π) designed to be risk-neutral:

Position 1: Short one option (Value = $-V$)

Position 2: Long Δ shares of the stock (Value = $(\partial V / \partial S) \times S$)

$$\text{Portfolio Value: } \Pi = -V + \frac{\partial V}{\partial S} S$$

The Cancellation

Now, we examine the change in the portfolio's value, $d\Pi = -dV + (\partial V / \partial S) dS$. By substituting the expressions for dV and dS , the random dW terms perfectly cancel out.

$$\begin{aligned} & -\sigma S \frac{\partial V}{\partial S} dW \\ & + \left(\frac{\partial V}{\partial S} \right) \times (\sigma S dW) \end{aligned}$$

The Critical Result

$$d\Pi = \left(-\frac{\partial V}{\partial t} - \frac{1}{2} \sigma^2 S^2 \left(\frac{\partial^2 V}{\partial S^2} \right) \right) dt$$

The stochastic term has vanished. The portfolio's change in value is now completely deterministic.

Derivation Step 4: The No-Arbitrage Argument

The Logic

In an efficient market, any portfolio with zero risk must earn the risk-free rate, r .

Therefore: $d\Pi = r\Pi dt$

The Final Step

We set our two expressions for $d\Pi$ equal:

$$\left(-\frac{\partial V}{\partial t} - \frac{1}{2}\sigma^2 S^2 \left(\frac{\partial^2 V}{\partial S^2} \right) \right) dt = r \left(-V + \frac{\partial V}{\partial S} S \right) dt$$

Rearranging the terms yields the Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

The Mystery Solved

Notice that μ , the expected return of the stock, has also vanished. The option price is independent of investors' expectations about the stock's performance. The price is determined only by observable, objective parameters.

The Solution: The Black-Scholes Pricing Formulas

While the PDE describes the relationship, Black and Scholes provided a direct closed-form solution for European options.

European Call Option Price (C)

$$C = S_0 N(d_1) - X e^{-rT} N(d_2)$$

European Put Option Price (P)

$$P = X e^{-rT} N(-d_2) - S_0 N(-d_1)$$

Where

d_1 Formula

$$d_1 = [\ln(S_0/X) + (r + 0.5\sigma^2)T] / \sigma\sqrt{T}$$

d_2 Formula

$$d_2 = d_1 - \sigma\sqrt{T}$$

S_0 : Current stock price

X : Strike price

T : Time to expiration (in years)

r : Risk-free interest rate

σ : Annualised volatility

$N(\cdot)$: Cumulative standard normal distribution function

The Formula in Action: A Worked Example

Scenario Parameters

Current Stock Price (S_0): **\$100**

Strike Price (X): **\$100**

Time to Expiration (T): **1 year**

Risk-Free Rate (r): **5%**

Volatility (σ): **20%**

1. Calculate d_1

$$d_1 = [\ln(100/100) + (0.05 + 0.5 \times 0.2^2) \times 1] / (0.2\sqrt{1}) = 0.35$$

2. Calculate d_2

$$d_2 = 0.35 - 0.2\sqrt{1} = 0.15$$

3. Find $N(d_1)$ and $N(d_2)$

$$N(0.35) \approx 0.637$$

$$N(0.15) \approx 0.560$$

4. Calculate Call Price (C)

$$C = (100 \times 0.637) - (100 \times e^{-0.05} \times 0.560)$$

$$C = 63.70 - 53.05 = \mathbf{\$10.65}$$

The at-the-money call option is worth approximately **\$10.65**.

Decoding the Formula: The Meaning of $N(d_1)$ and $N(d_2)$

$N(d_1)$: The Option's Delta

$$N(d_1) = \Delta = \frac{\partial C}{\partial S}$$

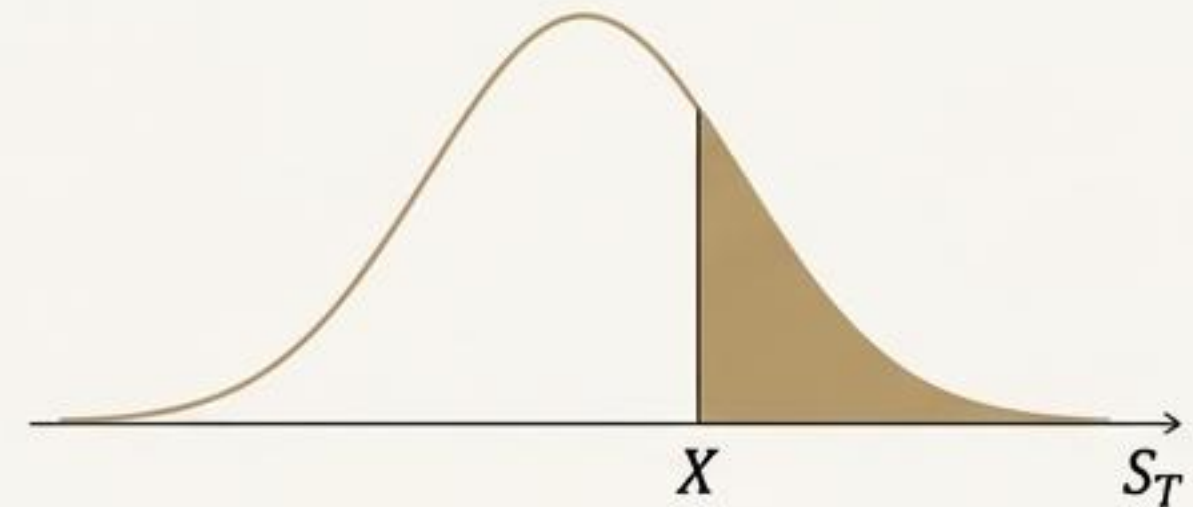
It is the **hedge ratio**. It represents the option's sensitivity to a change in the underlying stock price.

If $N(d_1) = 0.60$, the call price increases by £0.60 for every £1 increase in the stock price. To hedge a short call, one must buy 0.60 shares of the stock.

Varies from 0 (deep out-of-the-money) to 1 (deep in-the-money).

$N(d_2)$: The Probability of Exercise

It is the **risk-neutral probability** that the option will expire in-the-money (i.e., **Probability**($S_T > X$)).



The term $Xe^{-rT}N(d_2)$ is the present value of the expected payment of the strike price upon exercise.

$$C = \underbrace{[S_0 N(d_1)]}_{\text{Expected benefit of acquiring stock}} - \underbrace{[Xe^{-rT} N(d_2)]}_{\text{Present value of paying the strike price}}$$

The Trader's Dashboard: Managing Risk with The Greeks (Part 1)

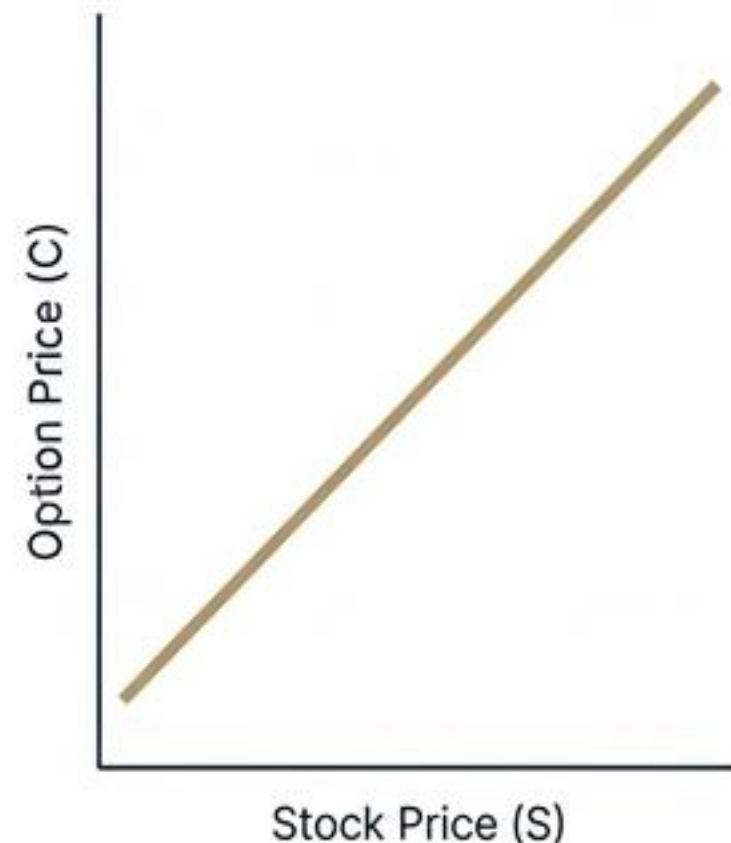
The Greeks are the partial derivatives of the option price, measuring sensitivity to different market parameters.

Delta (Δ): Price Sensitivity

$$\Delta_{\text{call}} = N(d_1)$$

Measures the rate of change of the option price with respect to a change in the underlying stock price. The primary primary measure of directional exposure.

- Delta-neutral hedging aims to eliminate directional risk.

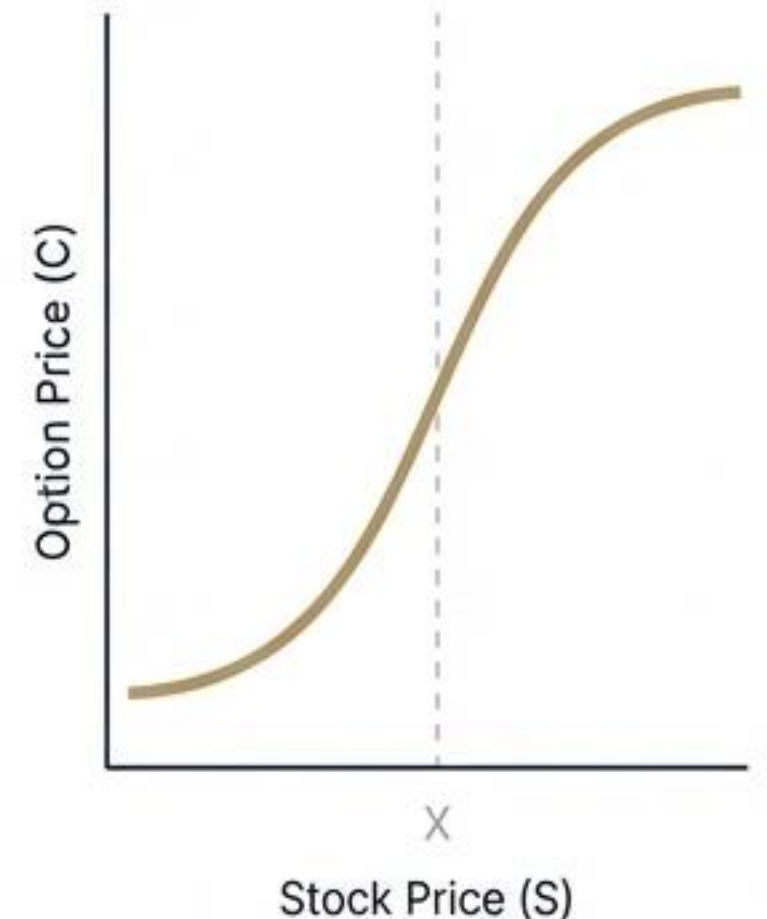


Gamma (Γ): Delta's Sensitivity

$$\Gamma = \frac{N'(d_1)}{S\sigma\sqrt{T}}$$

Measures the rate of change in Delta. It shows how much the hedge ratio itself will change as the stock price moves.

Highest for at-the-money options near expiration. Long gamma positions profit from large price moves (volatility), while short gamma positions are harmed by them.



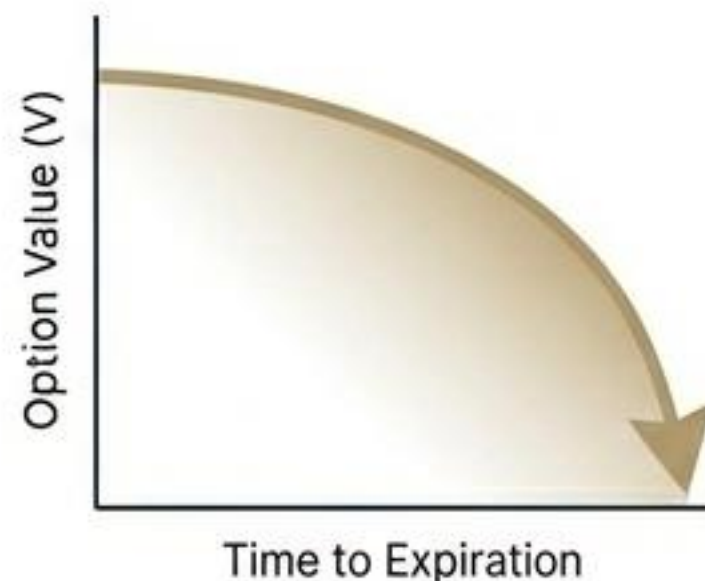
The Trader's Dashboard: Managing Risk with The Greeks (Part 2)

Theta (Θ): Time Decay Sensitivity

$$\Theta = -\frac{\partial V}{\partial t}$$

Measures the **loss** in option value per day as time passes, all else being equal. It is the "**cost of carry**" for the option holder.

Option sellers profit from theta decay, especially for near-expiration options.



Vega (ν): Volatility Sensitivity

$$\nu = \frac{\partial V}{\partial \sigma}$$

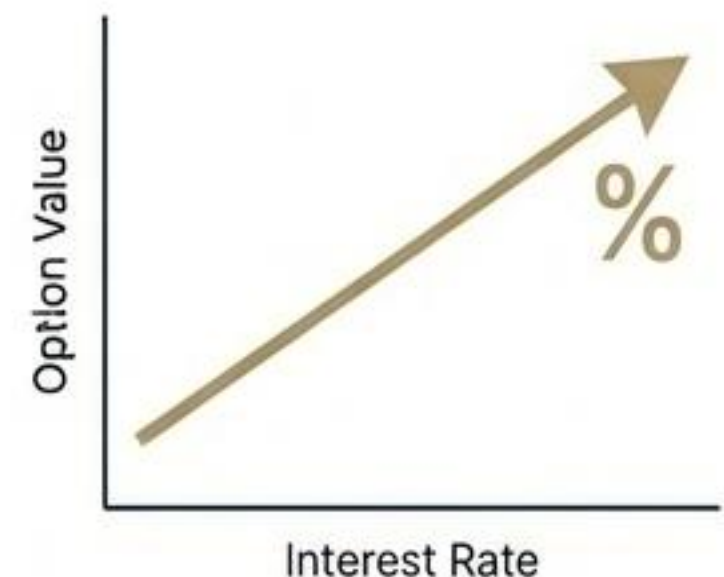
Measures sensitivity to a 1% change in implied volatility. It is not one of the original Greek letters.

Traders buy options (go "long vega") ahead of events like earnings announcements where volatility is expected to rise.



Rho (ρ): Interest Rate Sensitivity

Measures sensitivity to a 1% change in the **risk-free rate**. Generally the least significant Greek for short-dated options.



Synthesising Risk: P&L Attribution via Taylor Expansion

We can approximate the daily change in an option's value by summing the contributions from each Greek.

$$\Delta C \approx (\Delta \cdot \Delta S) + \left(\frac{1}{2}\Gamma \cdot (\Delta S)^2\right) + (\nu \cdot \Delta \sigma) + (\Theta \cdot \Delta t)$$

Practical Example: P&L Attribution for a Portfolio of 100 Long Calls

Initial Greeks

Delta=0.50, Gamma=0.5, Vega=0.2,
Theta=-\$0.01/day

Market Move

Stock +\$2, Volatility +2%, 1 day passes

P&L Breakdown

Delta P&L: $(0.50 \times 100) \times \$2 = +\$10,000$ (from direction)

Gamma P&L: $\frac{1}{2} \times 0.5 \times 100 \times (2)^2 = +\100 (from convexity)

Vega P&L: $0.2 \times 100 \times 2\% = +\40 (from volatility increase)

Theta P&L: $-0.01 \times 100 \times 1 = -\1 (from time decay)

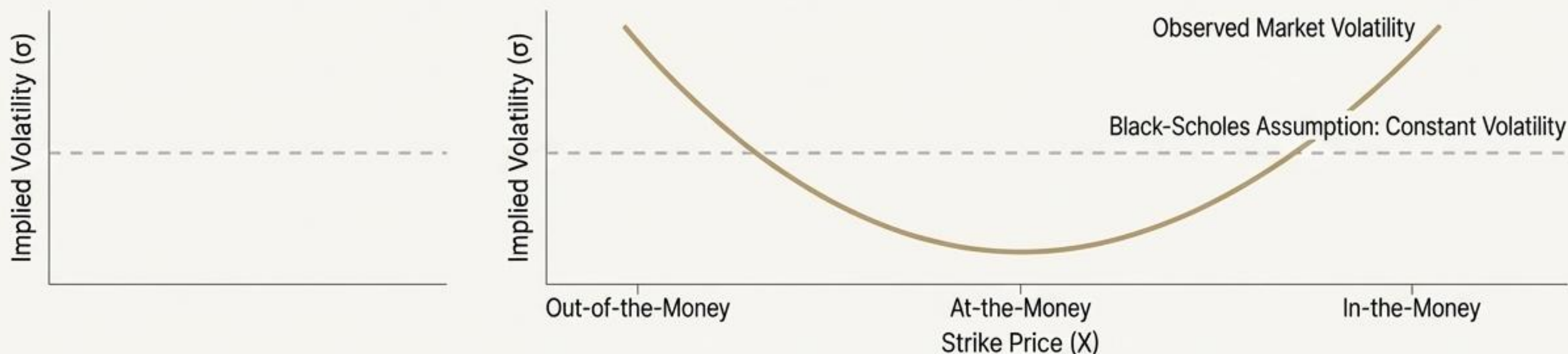
Total Estimated P&L $\approx$ \$10,139

This decomposition is essential for understanding the source of profits and losses in a derivatives portfolio.

The Reality Check: Unrealistic Assumptions and the Volatility Smile

The model's assumptions (constant volatility, no transaction costs, log-normal returns) do not hold in reality.

The Empirical Contradiction: The Volatility Smile



- In the real market, options with different strike prices but the same expiry have different implied volatilities.
 - **The Reason:** Real-world asset returns have "fat tails." Extreme events (crashes) are more common than a normal distribution predicts, making deep out-of-the-money puts more expensive.
- **The Paradox and The Practice:** Despite violating a core assumption, traders do not discard the model. Instead, they use the Black-Scholes formula "in reverse" to calculate the implied volatility for each option, creating a "volatility surface."
 - The model's framework is so robust that it becomes the tool used to map its own imperfections.

A Revolutionary Legacy: The Equation That Built an Industry

The Black-Scholes model was more than an academic exercise; it provided the foundational language and framework for the modern derivatives market.



Enabled Exchanges

Provided a consistent, theoretical basis for pricing, allowing options exchanges to flourish.



Standardised a Lexicon

Concepts like delta-hedging, gamma, and implied volatility became the industry standard for communicating and managing risk.



Launched Quantitative Finance

The mathematical rigour of the model became the blueprint for the entire discipline of financial engineering.

The work was awarded the 1997 Nobel Memorial Prize in Economic Sciences to Robert Merton and Myron Scholes (Fischer Black having passed away in 1995).

Despite its known limitations, the Black-Scholes framework remains the most widely used options pricing model globally—a testament to its fundamental correctness and enduring practical utility.